Shourya Sahdev

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Education

University of Illinois Urbana-Champaign *December 2024*
Master of Engineering in Mechanical Engineering
- **Coursework:** Finite Element Analysis, Applied Aerodynamics, Additive Manufacturing and Product Design, Design for Six Sigma, Experimental Stress Analysis, Carbon Capture and Storage, Electronics Cooling, Computational Mechanics

Delhi Technological University *June 2021*
Bachelor of Technology in Production and Industrial Engineering
- **Coursework:** Solid Mechanics, Fluid Mechanics, Thermodynamics, Robotics, Mechatronics, Quality Engineering

Experience

Process Engineering Assistant, Micro and Nano Technology Lab *June 2024 – Present*
- Streamlined inventory tracking for 20+ cleanroom consumables, minimizing manual checks and supply disruptions.
- Implemented chemical disposal protocols, lowering contamination risk and ensuring ISO-compliant cleanroom practices.

Mechanical Engineer (Capstone Project), SLB *January 2024 – May 2024*
- Engineered ring & solenoid dampers using viscoelastic PU and neoprene, ensuring manufacturability, reliability, and fit.
- Applied FEA in ANSYS with Dynamic Explicit models to evaluate linear & hyper-elastic materials for damper responses.
- Utilized Neo-Hookean & Holzapfel-Gasser-Ogden models to assess damper performance under high-impact shock loading.

Mechanical Engineer, Delhi Technological University *December 2021 – January 2023*
- Designed quadcopter and fixed-wing UAV frames for additive manufacturing, integrating motors, actuators, and electronics.
- Performed material analysis & structural FEA in ANSYS/Fusion 360; selected nylon for optimal strength-to-weight ratio and machinability, achieving 30% lighter quadcopter frame for FDM 3D printing and executed rapid prototyping to validate design.
- Built Python tool to compute Cl, Cd, and geometry; applied Matplotlib plots to optimize UAV for efficiency & range.
- Modeled UAV in OpenVSP and validated lift/drag coefficients with CFD in ANSYS Fluent to confirm aerodynamic design.
- Designed electromechanical cargo manipulator; created BOM, prototyped, and validated system performance in Simulink.

Co-Captain, Human Powered Vehicle Team, DTU *August 2018 – July 2021*
- Led a 20-member team through full-cycle vehicle development, managing timelines, reviews, and on-track testing.
- Engineered bike and tadpole frame geometries in SolidWorks, incorporating insights from vehicle dynamics, user-centric ergonomic feedback, and Whipple model insights, enabling the vehicle to reach 50 mph during hands-on testing.
- Applied DFMA principles and created GD&T-compliant drawings for efficient prototyping and assembly.
- Collaborated on design and integration of aerodynamic fairing, reducing frontal area by 40% and boosting performance.
- Validated structural integrity and optimized idler design using ANSYS FEA for composite analysis, topology optimization.
- Reduced carbon fiber fairing production costs by 60% by developing a mold manufacturing process using polystyrene.

Industrial Engineer Intern, Maruti Suzuki *June 2019 – July 2019*
- Performed Failure Mode Analysis to identify 7 failure modes in spindle assembly, reducing lead time and costs by 15%.
- Trained assembly workers on proper bearing installation techniques, minimizing premature breakdowns.

Projects

Single Phase Immersion Cooling Solution for EV Batteries *August 2024 – December 2024*
- Engineered single-phase immersion cooling model for EV packs; validated thermal profile with ANSYS FEA.
- Achieved 46% reduction in maximum battery temperature with flow immersion cooling compared to cold plate.

Design in Six Sigma *August 2024 – December 2024*
- Executed PFMEA and RCA across 4 case studies using DMAIC to identify failure modes and reduce DPMO by 40%.
- Performed Gage R&R and DOE in Minitab to validate measurement systems and improve Cpk and Ppk values by 60%.

Experimental Stress Analysis *January 2024 – May 2024*
- Conducted Beam Impact Testing of PVC pipes using strain gauges under quasi-static and dynamic loading.
- Tracked particles in soft solids via Python and TrackMate, quantifying strain fields, crack growth, fracture energy.
- Applied photoelasticity and DIC to horseshoe specimens, validating stress-strain fields against theoretical models.

Technical Skills

Software: SolidWorks, ANSYS(Static Structural, Fluent), Fusion360, AutoCAD, StarCCM, CATIA, Minitab, Simulink
Programming Language & Library: C, C++, Python, MATLAB, Numpy, Matplotlib, HTML, CSS, Arduino
Process: Mechanical Design, Vehicle Architecture, DFMA, GD&T, Rapid Prototyping, Tolerancing, Surfacing, RCA